

Newborough Warren

Hydrological Scenario Viewer

Technical Note

Version 1.0 — April 2026

Architecture, methods, implementation

Companion to: Hollingham (2026)
Journal of Hydrology: Regional Studies (submitted)

scenario_viewer.html — generated by 19_spatial_groundwater.py
https://newbroman.github.io/Newborough_Hydrology/scenario_viewer.html

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1. Introduction

This technical note documents the implementation of `scenario_viewer.html`, the browser-based scenario-exploration tool built as a companion to Hollingham (2026). It is intended for researchers who want to understand how the viewer is constructed, which methodological choices were made and why, and how the viewer can be reproduced, audited, or extended. For a practitioner-facing description of what the viewer does and how to use it, see the companion Reference Manual.

The viewer is generated by a single Python script, `19_spatial_groundwater.py`, which sits at the end of a nineteen-script analytical pipeline (scripts 01–19). The script consumes the products of earlier pipeline stages — per-well state-space model (SSM) coefficients, cluster assignments, specific yield estimates from the water-table fluctuation method, boundary-subsidy intercepts — together with LiDAR digital elevation data and site-boundary polygons, and produces two outputs: a self-contained HTML file (~770 KB) and a tabular CSV scenario summary (`19_scenario_summary.csv`) suitable for direct citation in the main paper (Section 4.9).

2. Architecture

The viewer is a single self-contained HTML file. All data, all code, and all rendering assets are embedded directly into the file; after the initial page load no server, no internet connection, and no external dependency (beyond the reader's browser and a one-time CDN fetch of Chart.js for the hydrograph panel) is required. This design is deliberate: it allows the viewer to be archived alongside the manuscript, hosted on any static file server including GitHub Pages, emailed as an attachment, or run offline from a local file for reproducibility verification.

Inside the HTML file, the embedded data structures are:

- **WELLS** — a JSON array of 71 well records, each carrying location (OSGB36 E, N), cluster membership, ground-surface elevation, pipe-top elevation, and the fitted $\beta_1/\beta_2/\beta_3$ state-space coefficients (scripts 01, 02, 03).
- **CLIMATE** — a JSON object holding the baseline seasonal and annual P and PET values for each cluster, plus the full 12-month climatological means used by the viewer's seasonal-slider mathematics.
- **POLYS** — a JSON object holding the site boundary and the Corsican pine plantation footprint as polygon vertex arrays; also the Llyn Rhos-ddu lake polygon.
- **HILLSHADE_B64** — a base64-encoded PNG of the LiDAR-derived greyscale hillshade (see Section 9).
- **DEM_GRID** — a 160×110 JSON grid of DEM elevations used for ridge masking in depth-below-surface mode (see Section 8).
- **SCEN** — the six scenario definitions (baseline, UKCP18 2050s, UKCP18 2080s, clearfell, broadleaf, thinning) as multipliers on the seasonal forcing terms.

The JavaScript inside the viewer performs all computation at the browser. Scenario selection and slider movement trigger re-evaluation of the per-well equilibrium Δh at every one of the 71 reference wells; the resulting point set is then interpolated to the map display via inverse-distance weighting. No round-trip to a server is involved, and the viewer remains responsive under continuous slider motion.

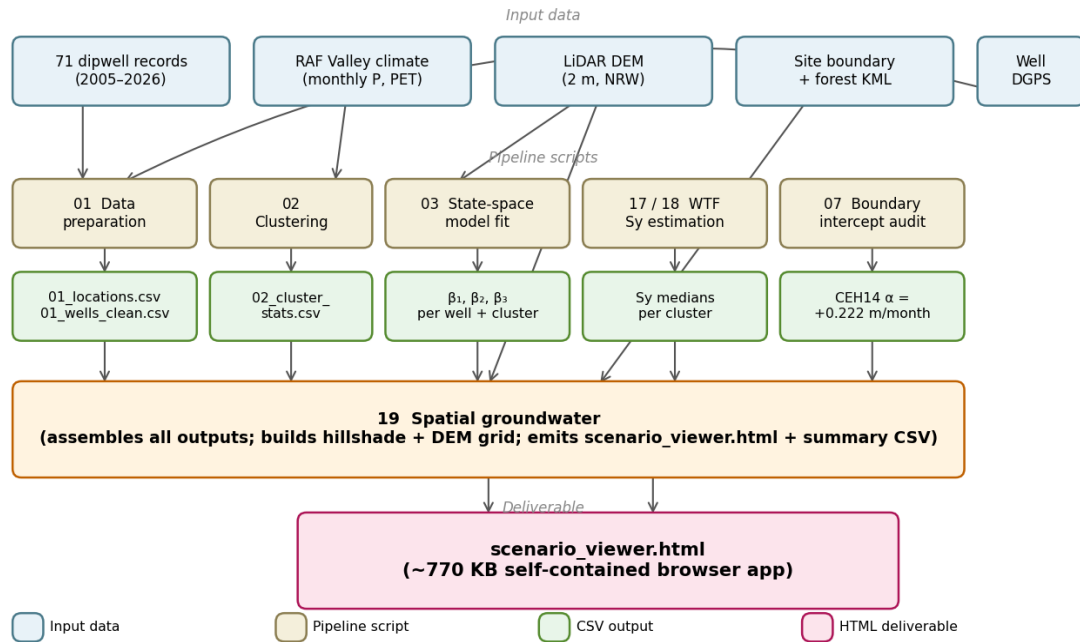


Figure 1. Pipeline architecture: input data (blue) feeds pipeline scripts (beige) whose CSV outputs (green) are all consumed by script 19 (orange), which emits the self-contained HTML viewer (pink).

3. Input data

The viewer inherits its inputs from the earlier stages of the pipeline. Table 1 documents which script produces which embedded payload, and which CSV file in the pipeline output tree is the proximate source for each.

Payload	Producing script	CSV source
WELLS array — well locations, cluster, elevations, β coefficients	01 (location, elevation); 02 (cluster); 03 (SSM fit)	01_locations.csv, 02_cluster_stats.csv, 03_master_data.csv
CLIMATE object — baseline seasonal and monthly P, PET means	01 (climate ingestion); 03 (cluster averages)	01_climate.csv, 03_regional_averages.csv
SCEN object — scenario perturbation multipliers	19 (hard-coded in script, referenced in Section 10)	—
HILLSHADE_B64 — base64 PNG basemap	19 (calls map_utils.load_dem_hillshade)	newborough_dem.tif (2 m LiDAR, NRW)
DEM_GRID — 160×110 elevation grid for ridge mask	19 (built from the same LiDAR DEM)	newborough_dem.tif
POLYS — site boundary, plantation, lake	19 (reads KML, applies viewer-extent clip)	site_boundary.kml, forest.kml, llyn_rhosddu.kml
Specific yield displays	17 / 18 (WTF method, cluster medians)	17_wtf_well_sy.csv

The 71-well count is the monitoring network supporting the state-space model fit; within this, 68 wells have sufficient record length to yield statistically significant β coefficients. Wells CEH3 (C5 Coastal) and CEH15 (forest slack floor) are excluded from interpolation throughout the viewer: CEH3 because it is sea-boundary dominated and produces unstable SSM fits, CEH15 because it represents forest-floor water-table dynamics governed by canopy throughfall rather than regional groundwater flow.

4. Coordinate system and map extent

All coordinates in the viewer are in the Ordnance Survey of Great Britain 1936 projection (OSGB36 / EPSG:27700). This is the reference system of the UK national grid and is used uniformly across the pipeline — the LiDAR DEM, the KML polygons, and the dipwell DGPS survey are all in OSGB36. No reprojection is performed at the viewer, which simplifies both the pipeline and the viewer's internal arithmetic: distances are in metres, eastings and northings are interchangeable with map coordinates, and the aspect ratio of the displayed map preserves the geographic shape of the site to within the tolerance of the PNG hillshade sampling (Section 9).

The viewer extent is:

- Easting: 240200 m to 243700 m (3500 m span)
- Northing: 362400 m to 364800 m (2400 m span)

The extent is defined in `19_spatial_groundwater.py` as a single set of module-level constants (`VIEWER_EMIN`, `VIEWER_EMAX`, `VIEWER_NMIN`, `VIEWER_NMAX`) and referenced consistently by the hillshade builder, the DEM grid builder, the forest-polygon clipping rectangle, and the JavaScript template. This single-source-of-truth convention eliminates the drift that would otherwise occur if any of these five locations were updated independently.

The canvas-to-world transformation is a simple affine map. The HTML canvas has its origin in the top-left corner with the y-axis increasing downward; the OSGB36 grid has its origin in the bottom-left corner with northing increasing upward. The transformation applied at every rendering call is:

$$x_{\text{canvas}} = (E - EMIN) / (EMAX - EMIN) \cdot W$$

$$y_{\text{canvas}} = H - (N - NMIN) / (NMAX - NMIN) \cdot H$$

where W and H are the canvas width and height in pixels (640 and 440 respectively, fixed in v2.2.0). The inverse transformation is applied when the user hovers over the map to retrieve the OSGB36 coordinate under the cursor.

5. State-space model

The viewer is built on the state-space model (SSM) of Hollingham (2026). At each dipwell the monthly change in water-table elevation is modelled as:

$$\Delta h(t) = \beta_1 \cdot P_{eff}(t) - \beta_2 \cdot PET(t) - \beta_3 \cdot h(t-1)$$

where P_{eff} is effective rainfall reaching the water table (after canopy interception, if applicable), PET is Thornthwaite–Mather potential evapotranspiration, and $h(t-1)$ is the well's depth-below-ground in the previous month. The β coefficients are fitted independently at each of the 68 reference wells using ordinary least squares, with sign conventions enforced at the fit stage so that all three β values are positive magnitudes in the reported output. The fit is performed by script 03, which writes per-well coefficients to 03_master_data.csv; cluster-level aggregates used for the Section 4.9 analysis are in 03_03_cluster_mechanistic_coefficients.csv.

5.1 Equilibrium response

The viewer operates in equilibrium rather than dynamically. Setting $\Delta h = 0$ in the SSM equation and solving for the steady-state head under scenario forcing gives:

$$h_{eq} = (\beta_1 \cdot P_{eff} - \beta_2 \cdot PET) / \beta_3$$

The change in equilibrium head between the baseline and a scenario is then a simple subtraction. The internal drainage term β_3 cancels between scenario and baseline expressions in the Δh calculation, as long as it is held constant (which it is in all scenarios except those that explicitly modify C4 Forest β values). Specific yield does not appear in this equation; it enters only the dynamical formulation of the SSM, not the equilibrium, because the fitted β coefficients already embed the aquifer storage response through the fit to observed heads. The viewer reports Sy in the summary panel for practitioner context, and the Sy slider is a display scaling only.

5.2 Scenario application

Two scenario classes behave differently in the SSM:

- Climate scenarios (UKCP18 2050s, UKCP18 2080s, and any user-driven slider perturbation) modify the P and PET forcing terms uniformly at all 71 wells. The resulting Δh map is spatially continuous and reflects the inherited β -coefficient structure across clusters.
- Forest-management scenarios (clearfell, broadleaf, thinning) modify the canopy interception fraction and (for clearfell) the β_2 multiplier at only the nine C4 Forest cluster wells. Open-dune, Eastern-Block, and Coastal cluster wells see no parameter change. The response is therefore spatially confined to the C4 zone and its immediate interpolated surroundings.

6. Spatial interpolation

The viewer uses inverse-distance weighting (IDW) with fixed parameters for all map rendering, while the static figures in the main paper (Figures 31 and 32) use Delaunay triangulation with linear barycentric weighting. The two methods are used for different purposes and the choice is deliberate.

6.1 The viewer's IDW kernel

At every grid cell (x, y) , the interpolated value $f(x, y)$ is computed as a weighted mean over the k nearest sample wells:

$$f(x, y) = \frac{\sum_{i=1}^k w_i \cdot f_i}{\sum_{i=1}^k w_i}$$

with weights:

$$w_i = 1 / \max(d_i, d_{\min})^p$$

where d_i is the Euclidean distance from the grid cell to sample well i , $d_{\min}$ is a small distance floor (10 m) to prevent singularity at well locations, and p is the distance exponent. The viewer uses $p = 1$ and $k = 8$ (eight nearest neighbours). An additional exact-value shortcut returns the well's own value when d_i is less than 5 m, so that sample locations are honoured exactly.

6.2 Why power 1 and not the conventional power 2

A conventional IDW exponent is $p = 2$, which weights distance more aggressively and produces a sharper response around sample points. During development, $p = 2$ was tested first but produced unacceptable bullseye halo artefacts around individual wells on the Δh surface: the sharp fall-off meant that each well acted as a discrete point source at its grid cell, surrounded by a gradient visibly discontinuous from its neighbours. $p = 1$ spreads the weight more smoothly and eliminates the halo effect, at the cost of a visually softer transition at well locations. The softer transition is acceptable for the viewer because the purpose is scenario visualisation rather than fine-scale quantification.

6.3 Why IDW in the viewer but triangulation-linear in the figures

The static figures in the main paper are rendered once and do not need to respond to slider input. Delaunay triangulation with linear barycentric weighting produces a cleaner, more mathematically-principled surface than IDW, but is computationally more expensive and produces a non-smooth gradient at triangle edges that is acceptable in a static figure but visually distracting in an interactive view where the surface updates in real time. IDW at power 1 with a fixed neighbour count is fast, smooth, and predictable under slider-driven re-evaluation. Using different methods for the two outputs — and documenting the fact — is a pragmatic compromise; the alternative, using IDW for the published figures, would compromise the static output for no benefit.

A consequence of the method difference is that the viewer extrapolates beyond the convex hull of the wells (using the IDW kernel), whereas the triangulation-linear figures correctly leave the extrapolated region blank. This is most visible near the coastal margin and in the south-east corner. Values in the extrapolated zone should be treated as indicative only.

7. Depth-below-surface mode

The depth-below-surface display mode is a viewer-specific visualisation not reproduced in the main paper's static figures. It is computed per grid cell as:

$$d(E, N) = h_{DEM}(E, N) - h_{WT}(E, N)$$

where h_{DEM} is the bilinearly-interpolated LiDAR ground-surface elevation at the cell centre and h_{WT} is the IDW-interpolated scenario water-table elevation at the same cell. Both quantities are in metres above ordnance datum, so their difference is depth below surface in metres. The bilinear DEM lookup is performed against the embedded 160×110 DEM_GRID (see Section 8 for the grid construction); DEM values at cells outside the grid are set to NODATA and the corresponding depth is reported as undefined.

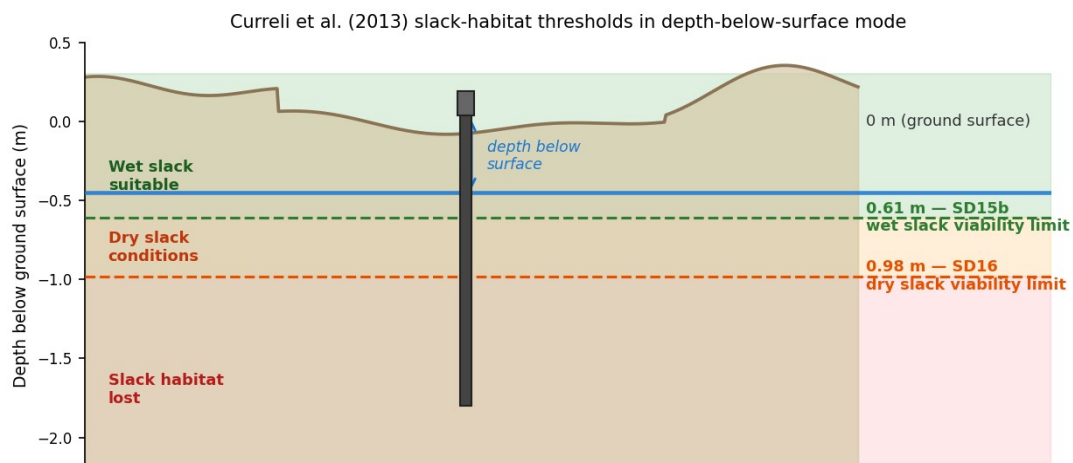


Figure 2. Depth-below-surface mode uses a custom colour scale anchored on the Curreli et al. (2013) slack-vegetation thresholds (SD15b wet slack: 0.61 m; SD16 dry slack: 0.98 m). The colour-scale anchoring allows scenario outputs to be read directly against ecological thresholds without arithmetic.

The colour scale is implemented in the viewer as a piecewise-linear ramp with four anchor colours: light blue at 0.10 m (winter flooding limit), green at 0.61 m, orange at 0.98 m, and deep red at 1.50 m. Explicit tick marks appear on the legend at the SD15b and SD16 threshold depths. The scale is independent of the Δh and Absolute head scales and is not toggled by the auto-rescaling logic that operates in those modes.

8. Dune-ridge masking

In depth-below-surface mode the viewer applies a mask to grid cells sitting on dune ridges. Without the mask, such cells would show the water table as lying several metres below ground and render as deep-red depth values — but this is a consequence of the ridge height, not of any hydrological feature, and would obscure the ecologically relevant slack-floor response. The mask is applied exclusively in depth mode; in Δh and Absolute head mode no masking is applied because the water table is a continuous surface whose elevation is well-defined everywhere.

8.1 Construction of the DEM grid

The embedded DEM_GRID is a 160-column $\times$ 110-row array of mean elevations, sampled from the Natural Resources Wales 2 m LiDAR DEM by block-averaging within each grid cell. The array is stored in row-major order in a JSON structure with explicit EMIN/EMAX/NMIN/NMAX bounds and a NODATA sentinel of -9999. The cell size is approximately 22 m on both axes, which safely exceeds the viewer's pixel-block rendering step (≈ 15 px) and keeps the embedded JSON payload under ~ 80 KB.

8.2 The ridge-mask test

For each display cell the viewer computes two quantities: the DEM elevation at the cell centre (via bilinear interpolation of DEM_GRID), and the well-plane elevation computed by the same IDW kernel used for the water-table surface but applied to the well ground elevations. A cell is masked in depth mode when:

$$h_{\{DEM\}}(E, N) - h_{\{wellplane\}}(E, N) > 1.0 \text{ m}$$

i.e. when the cell sits more than one metre above the surrounding slack-floor plane. The 1.0 m threshold matches the ridge-mask convention applied in the static figures (map_utils.add_idw_surface default). The mask is user-toggleable from the viewer's MAP panel; disabling it reveals the unmasked depth field, which is occasionally useful for diagnostic inspection but should not be used for ecological interpretation.

9. Hillshade basemap

The map background is a greyscale hillshade rendered from the 2 m LiDAR DEM (Natural Resources Wales, Open Government Licence v3.0). The hillshade is generated during pipeline execution, embedded directly in the HTML as a base64-encoded PNG, and stretched to fill the canvas by the viewer at every rendering call.

9.1 Rendering parameters

Parameter	Value	Rationale
Illumination azimuth	315° (NW)	Standard convention for visually natural shading in the Northern Hemisphere
Illumination altitude	35°	Moderate sun angle — emphasises topography without crushing shadow detail
Vertical exaggeration	3×	Compensates for the shallow relief of a coastal dune system
Output size	1100 × 750 px	Aspect 1.467 — matches the viewer extent aspect of 3500/2400 = 1.458
Greyscale quantisation	32 levels	PNG compression ratio; visually indistinguishable from 256-level at rendered size
Final payload	~200–300 KB base64	Acceptable contribution to a ~770 KB total HTML
Render opacity	35% under surface	Matches the static figure convention

The hillshade and the DEM grid are generated from the same source raster, with consistent extent and sampling, so that the ridge mask (Section 8) and the basemap align pixel-for-pixel at the viewer. The PNG hillshade is stretched to the canvas dimensions at every frame via the HTML Canvas `drawImage` method; the geographic extent covered by the PNG is identical to the viewer extent, so the stretch is correctly geo-referenced by construction.

10. UKCP18 seasonal parameterisation

The two UKCP18 scenarios in the viewer (2050s and 2080s) implement the RCP8.5 central-estimate (50th-percentile) projections for Wales, taken from the UKCP18 Regional 12 km product (Met Office, 2018) and informed by the CHES-SCAPE bias-corrected projections of Robinson et al. (2023). The perturbations are applied separately to the winter (November–March) and summer (May–September) seasons because the seasonal asymmetry — wetter winters, drier summers, and substantially higher summer PET — is the principal structural feature of the UKCP18 projection for the region and is responsible for the viewer’s main finding: that annual equilibrium Δh is dominated by the summer signal.

10.1 Scenario perturbations

Scenario	Winter P	Summer P	Winter PET	Summer PET
Baseline	1.00×	1.00×	1.00×	1.00×
UKCP18 2050s (2040–2069)	1.10×	0.85×	1.05×	1.20×
UKCP18 2080s (2070–2099)	1.20×	0.70×	1.10×	1.35×

These perturbations feed directly into the SSM equilibrium calculation of Section 5 by modifying the P_{eff} and PET terms. The annual Δh reported in the viewer summary panel is computed as:

$$\% \Delta h_{\text{annual}} = 0.5 \cdot (\% \Delta h_{\text{winter}} + \% \Delta h_{\text{summer}})$$

This equal-weighting aggregation preserves the seasonal asymmetry; an annual-mean forcing (averaging the perturbations before running the SSM) would average out the contrasting winter and summer responses and obscure the dominant summer signal.

10.2 Manual exploration via the seasonal sliders

The four seasonal sliders (Winter P, Summer P, Winter PET, Summer PET) allow independent perturbation of each forcing term across the range 0.5×–1.5× of baseline. This range is chosen to bracket the UKCP18 end-century probabilistic projections for Wales under RCP8.5 at the 10th and 90th percentiles, while staying within the linear steady-state domain of the fitted SSM. Selecting a scenario preset sets the sliders to the scenario values; subsequent slider movement perturbs the active scenario without overwriting the preset.

11. Performance and slider-driven re-interpolation

The viewer is designed for continuous interactive use. Slider movement and scenario selection trigger immediate re-computation of the per-well equilibrium Δh , re-interpolation to the map grid, and re-rendering of the display. The design target is that interactive updates should feel instantaneous on any modern browser and on reasonable mobile hardware.

Two design choices support this responsiveness. First, the per-well β coefficients are precomputed in the pipeline and embedded as flat JSON; at the viewer, applying a scenario reduces to 71 independent arithmetic operations per slider update — essentially free.

Second, the IDW interpolation runs in pixel blocks (≈ 15 px) rather than per-pixel, giving a 225 $\times$ reduction in per-frame computation at no loss of visual quality (because the 15 px block size is well below the spatial resolution of the Δh field at inter-well distances of 100–300 m).

The DEM bilinear lookup (Section 7) and the well-plane IDW (Section 8.2) add a modest cost in depth-below-surface mode; the ridge-mask test in particular is executed per display cell. Measurements during development showed that depth-mode rendering is approximately 1.4 $\times$ slower than Δh mode on the same hardware, still well within the target for interactive use.

12. File size, deployment, and browser compatibility

The viewer is a single HTML file. Total size is approximately 770 KB, broken down approximately as follows:

Component	Size (KB, approx)	Notes
HILLSHADE_B64 (hillshade PNG)	220	Largest single component
DEM_GRID (JSON array)	75	160×110 elevation values
WELLS (JSON array)	55	71 wells × β coefficients + metadata
CLIMATE, POLYS, SCEN (JSON)	25	Small structured objects
JavaScript application code	40	Inline in the HTML, not externalised
HTML structure + CSS	15	Page chrome
External dependency (cached)	340 one-off	Chart.js loaded from cdnjs.cloudflare.com

The external Chart.js dependency is the only runtime network requirement. On first load the file is fetched from cdnjs.cloudflare.com (approximately 340 KB); subsequent visits use the browser cache. If the viewer is used in an offline setting, Chart.js can be vendored into the HTML by inlining — at a one-time edit to 19_spatial_groundwater.py — for a true zero-network deployment.

12.1 Browser compatibility

The viewer uses vanilla ES6 JavaScript and the HTML Canvas API. It has been tested on current versions of Chrome, Firefox, Safari, and Edge, on desktop (Windows, macOS, Linux) and mobile (iOS, Android). Internet Explorer is not supported. The canvas has a fixed pixel size of 640 × 440; on viewports narrower than 640 px the wrap element shrinks via a CSS max-width:100% rule and the canvas follows, preserving the aspect ratio.

13. Code and data availability

The full analytical pipeline (scripts 01–19, including 19_spatial_groundwater.py which builds the viewer) is archived on GitHub at https://github.com/newbroman/Newborough_Hydrology. The GitHub Pages site at https://newbroman.github.io/Newborough_Hydrology/ hosts the published scenario_viewer.html alongside this technical note and the reference manual.

The monitoring data underlying the SSM fit and the cluster analysis were independently collected and maintained by the author; the climate record is from the UK Met Office MIDAS archive (RAF Valley station) under Open Government Licence; the LiDAR DEM is from Natural Resources Wales under Open Government Licence v3.0. A permanent archive of the pipeline and the scenario_viewer.html is available on Zenodo under DOI <https://doi.org/10.5281/zenodo.19567643>.

The viewer is distributed under the MIT licence. It is an exploratory scenario-visualisation tool for research communication and conservation management planning, and is not intended for engineering or regulatory decisions.

14. References

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